Ancient pollen shows how fast the land changed around ice age footprints

From: Popular Science Mined: 2026-10-17 5 min read

Some of the oldest known human footprints in North America are pressed into a dry **lakebed** at White Sands National Park in New Mexico. A new study adds more evidence that they were made during the last **ice age**, about 23,500 to 21,700 years ago.

Archaeologists found the footprints in 2006. For a long time, many researchers thought the first people arrived in North America about 16,000 to 12,000 years ago. They believed people crossed **Beringia**, land that once connected Alaska to Russia. Footprints about 23,000 years old would mean people were living here at least 7,000 years earlier than expected, so scientists have debated their age for years.

For the new study, researchers from the University of Arizona worked with the U.S. Geological Survey's Radiocarbon Laboratory. They examined hundreds of thousands of tiny ancient **pollen** grains taken from the tracks. They combined that with chemical clues in the mud. The results were published in the journal Science Advances.

The clues showed what the area's plants, rainfall, temperature and dryness were like. They also revealed that the people who made the footprints lived through big changes. Within just a few decades, nearby evergreen forests turned into plains covered with oak and sage. A giant lake called Lake Oterio rose and shrank for about 1,500 years before it dried up completely. These changes closely matched climate changes happening around the world at the time.

"Throughout that whole sequence, people were walking there, leaving their tracks alongside those of mammoths, giant sloths and dire wolves," said University of Arizona **geoscientist** Jessica Tierney.

Word Treasure

ice age -- An ice age is a long period when Earth was much colder and huge sheets of ice covered large areas.

archaeologists -- Archaeologists are scientists who study how people lived long ago by examining what they left behind.

lakebed -- A lakebed is the ground at the bottom of a lake, which can be seen when the water dries up.

pollen -- Pollen is a fine powder made by plants that can last for thousands of years in mud.

Beringia -- Beringia was a stretch of land that once connected Alaska to Russia.

geoscientist -- A geoscientist is a scientist who studies Earth, including its rocks, water and climate.

Background you need

During the last ice age, so much water was frozen in ice sheets that sea levels were lower and land connected Alaska to Russia.

Many scientists long thought the first people reached North America by crossing that land, called Beringia, near the end of the ice age.

Footprints pressed into soft mud can harden and be buried, preserving them for thousands of years.

Pollen grains from plants survive in old mud, so scientists can use them to learn which plants grew nearby and what the climate was like.

Break it down

LEAD: A new study supports that White Sands footprints were made during the last ice age.
KEY EVENT: Pollen analysis published in Science Advances
+- WHO: University of Arizona researchers with the U.S. Geological Survey's Radiocarbon Laboratory
+- WHERE: A dry lakebed in White Sands National Park, New Mexico
BACKGROUND: Many thought people arrived 16,000 to 12,000 years ago via Beringia
EVIDENCE: Pollen and chemical clues show plants, rainfall, temperature and dryness
FINDINGS: Forests became oak and sage plains within decades; Lake Oterio dried up after about 1,500 years
QUOTE: Jessica Tierney says people walked alongside mammoths, giant sloths and dire wolves

Why it matters

The footprints help answer when people first reached the Americas and show how early humans lived through fast climate changes.

Test what you remember

1. Where are the footprints found?

(A) Denali in Alaska
(B) The Grand Canyon in Arizona
(C) Yellowstone National Park
(D) White Sands National Park in New Mexico

2. About how old does the new study say the footprints are?

(A) About 12,000 years
(B) About 2,000 years
(C) About 1 million years
(D) About 23,500 to 21,700 years

3. What did researchers examine from the tracks?

(A) Dinosaur bones
(B) Ice cores
(C) Hundreds of thousands of tiny pollen grains
(D) Stone tools

4. What happened to the nearby forests within a few decades?

(A) They froze into glaciers
(B) They grew taller
(C) They became a lake
(D) They turned into plains covered with oak and sage

5. Why have scientists debated the footprints' age?

(A) They were found in a museum
(B) The footprints belong to animals
(C) An age near 23,000 years would mean people arrived much earlier than many thought
(D) The park was closed

6. Which animals left tracks alongside the people, according to Jessica Tierney?

(A) Mammoths, giant sloths and dire wolves
(B) Dinosaurs and pterosaurs
(C) Polar bears and seals
(D) Horses, cows and sheep

Answer key (try the quiz first!): 1: D 2: D 3: C 4: D 5: C 6: A

Questions to think about

MY THOUGHTS (at least 20 words):